Problem Set 4 — Confidence Intervals & Hypothesis Testing

VCE Specialist Mathematics 3&4 · Chapter 16 · Weeks 5–6

Due: start of Week 7 (SAC week). Answers at the back — attempt everything first.

Name: ____________________ **Due:** Monday 24 August

Method reminder: CI: $\bar{x} \pm z\,\sigma/\sqrt{n}$. Test: state $H_0 : \mu = \mu_0$ and H_1 *from the wording*, compute $z = \dfrac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$, convert to a p-value (double it for two-tail), compare with α, conclude *in context*. Never "accept H_0".

Section A — Confidence intervals

A1. A sample of 64 components has mean lifetime $\bar{x} = 82$ hours; the population standard deviation is $\sigma = 8$ hours.

a.) Find a 95% confidence interval for μ. *(2 marks)*

b.) Find a 99% confidence interval for μ. *(2 marks)*

c.) Explain carefully what "95% confident" means. *(2 marks)*

A2. Measurements have $\sigma = 10$. What sample size guarantees a 95% confidence interval for μ of width at most 4? *(3 marks)*

Section B — Hypothesis tests

B1. A cereal filling process is set to $\mu = 120$ g with $\sigma = 15$ g. After a maintenance issue, a sample of 36 boxes has $\bar{x} = 114$ g. Test, at the 5% level, whether the mean fill has *decreased*. Show all five steps: hypotheses, distribution of $\bar{X}$ under H_0, test statistic, p-value, conclusion in context. *(6 marks)*

B2. Reaction times on a standard task have $\mu = 40$ ms, $\sigma = 6$ ms. Under new lighting, 25 trials give $\bar{x} = 42.1$ ms. Test at the 5% level whether the mean has *changed*. *(5 marks)*

B3. For the data in B2, compute a 95% confidence interval for μ, and explain how it is consistent with your test conclusion. *(3 marks)*

Section C — Errors

C1. For the test in B1:

a.) Describe a Type I error in context, and state its probability. *(2 marks)*

b.) Describe a Type II error in context. *(2 marks)*

C2. (Extension) Quality scores are $N(\mu,\ 10^2)$; we test $H_0 : \mu = 100$ against $H_1 : \mu > 100$ with $n = 25$ at $\alpha = 0.05$.

a.) Show that H_0 is rejected exactly when $\bar{x} \geq 103.29$. *(2 marks)*

b.) If in truth $\mu = 105$, find the probability of a Type II error. *(3 marks)*

Answers

A1. a.) $82 \pm 1.96(1) = (80.04,\ 83.96)$. b.) $82 \pm 2.576 = (79.42,\ 84.58)$. c.) Under repeated sampling, about 95% of intervals built this way contain μ — a fixed interval either contains μ or not.

A2. $2(1.96)\dfrac{10}{\sqrt{n}} \le 4 \Rightarrow \sqrt{n} \ge 9.8 \Rightarrow n \ge 96.04$: take $n = 97$.

B1. $H_0 : \mu = 120$, $H_1 : \mu < 120$. Under H_0, $\bar{X} \sim N(120,\ 2.5^2)$. $z = \dfrac{114 - 120}{2.5} = -2.4$; $p = P(Z \le -2.4) \approx 0.0082 < 0.05$: reject H_0 — evidence the mean fill has decreased.

B2. $H_0 : \mu = 40$, $H_1 : \mu \ne 40$. $z = \dfrac{42.1 - 40}{1.2} = 1.75$; $p = 2P(Z \ge 1.75) \approx 0.080 > 0.05$: do not reject H_0 — insufficient evidence of a change.

B3. $42.1 \pm 1.96(1.2) = (39.75,\ 44.45)$; it contains 40, matching "do not reject" at the 5% level.

C1. a.) Concluding the fill has decreased when it hasn't (unnecessary recalibration); $P = \alpha = 0.05$. b.) Failing to detect a genuine decrease (customers get underfilled boxes).

C2. a.) Reject when $z \ge 1.645$: $\bar{x} \ge 100 + 1.645(2) = 103.29$. b.) $P(\bar{X} < 103.29 \mid \mu = 105) = P\left(Z < \dfrac{103.29 - 105}{2}\right) = P(Z < -0.855) \approx 0.196$.